AIR UNIVERSITY
DEPARTMENT OF MECHATRONICS ENGINEERING
SEMESTER: SPRING 2026
Computer Programming
CE-112

Assignment-2

Instructor: Umer Farooq **Due Date:** 7th May 2026
Soft Copy Time :10:00 PM

Name: **Ashhad** **Name in ASCII (HEX)** **41 73 68 68 61 64**
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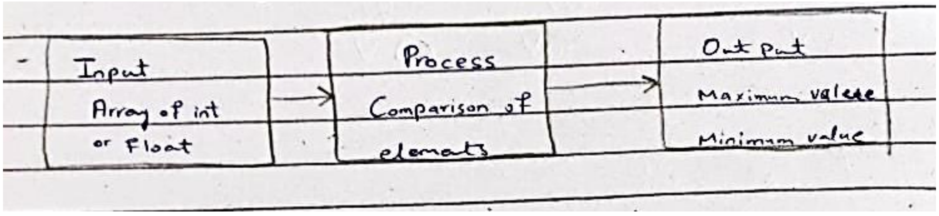
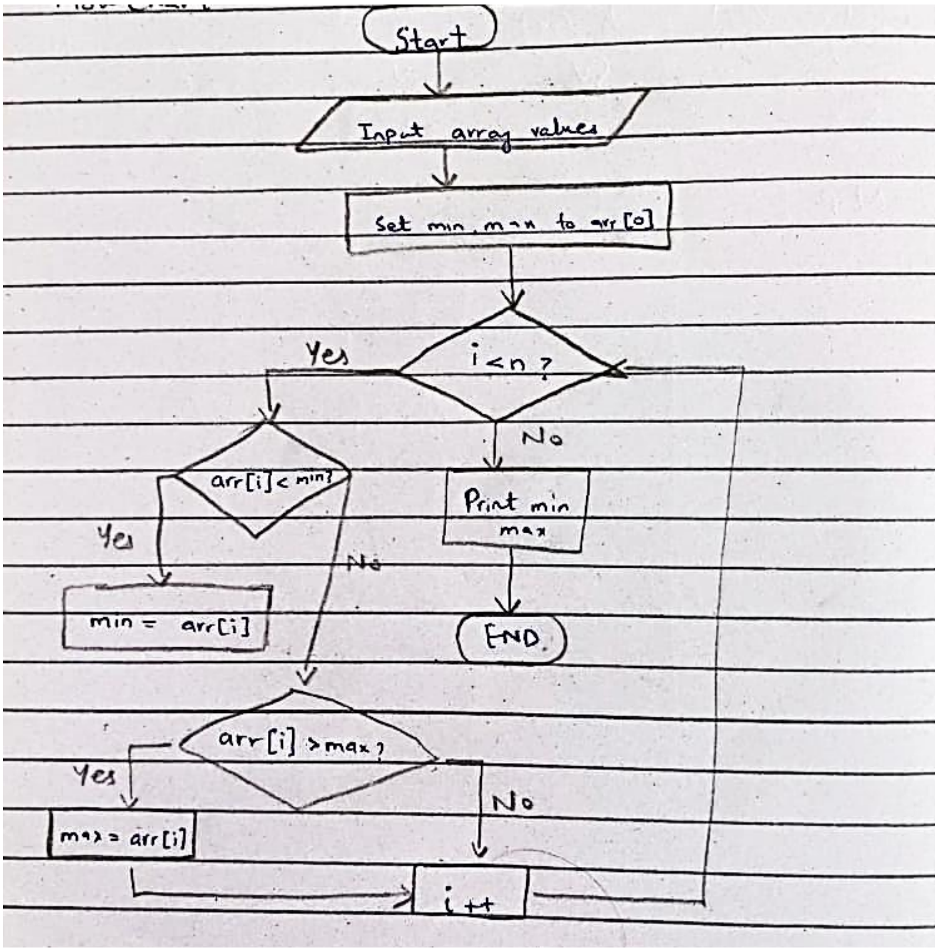
Question No	Write Status Complete/Not complete and issue that you faced	Remarks
1	Complete (no issue)	Calculates extremes for array of any size.
2	Complete (no issue)	Measures sample standard deviation of given data set.
3	Complete (no issue)	Multiply 3x3 matrices code can be edited to be more flexible for other dimensions.
4	Complete (no issue)	Counts number of different characters in any string.
5	Complete (no issue)	Calculates mean median mode of double precision data set.

Q1: Write and test the following function that returns through its reference parameters both the maximum and the minimum values stored in an array: getExtremes (float &min, float &max, float a[], int n);

Block Diagram



Flowchart



Code

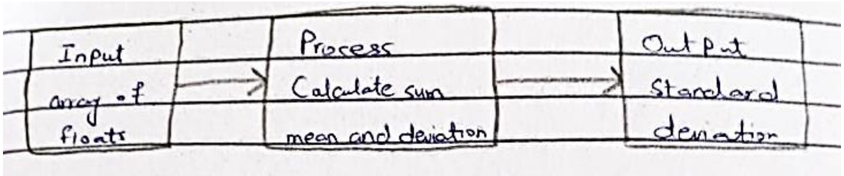
```
#include <iostream>
#include <cstdlib.h>

using namespace std;
// Function to find max and min element
void getExtremes (float &min, float &max, float arr[]
, int n) {
    min = arr [0];    // setting min, max to first element
    max = arr [0];
    for ( int i = 1; i < n;  i++) {  // array loop
        if (arr [i]  < min)   // updating min
            min = arr [i];
        if ( arr [i]  >  max)  // updating  max
            max  =  arr [i];
        }
    }
    // main function
int main () {
float min , max;   // variables  for results
float arr [] = { ———————  elements  —— }; // array
int n = size of (arr) / size of (arr [0]);  // size of arr
get Extremes ( min, max, arr, n);   // function call
cout << "Min: "  <<  min <<  endl << "Max: " << max << endl;
systen ( "pause");   // suspend process
return 0;
}
```

Output

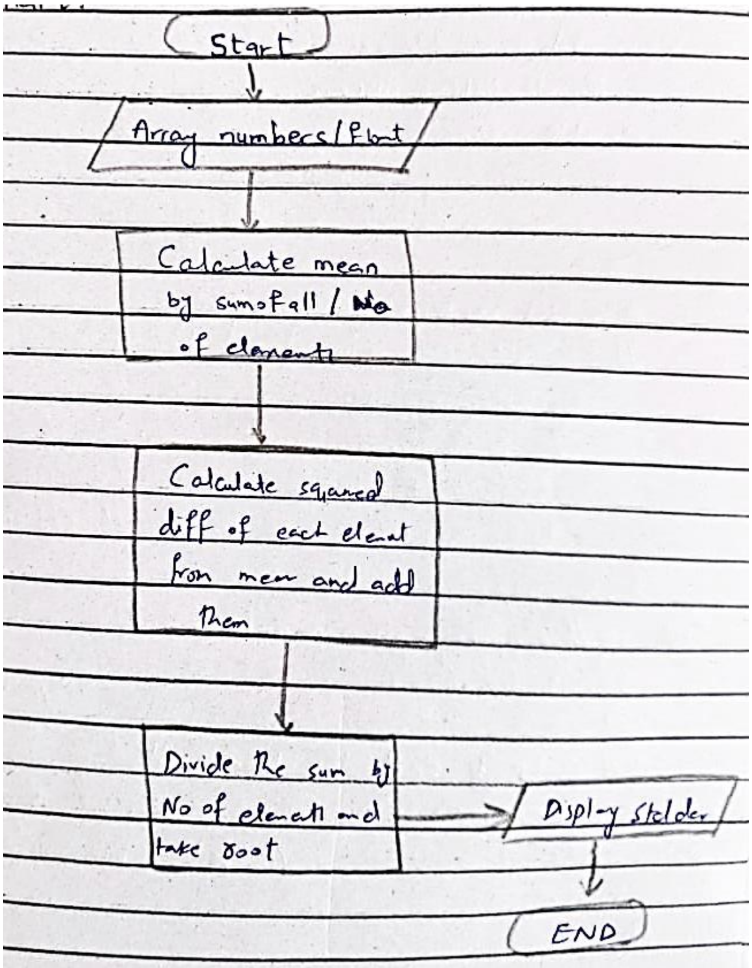
```
Min: 0.0528
Max: 0.9325
Press any key to continue . . .
```

Q2: Write and test the following function: double stdev(double x[], int n);

Block Diagram



Flowchart



Code

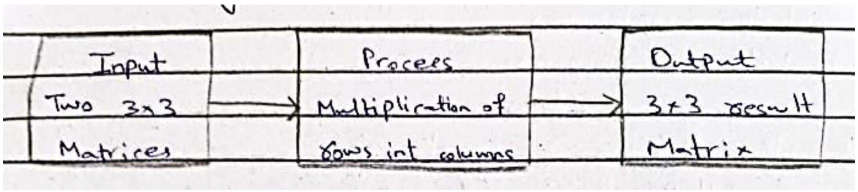
```
#include <iostream>
#include <cstdlib.h>
#include <cmath>

using namespace std;

double stdev(double x[], int n){
    //Calculating mean
    double sum = 0;
    for (int i = 0; i<n; i++){
        sum += x[i];
    }
    double mean = sum/n;
    // Calculating standard deviation
    double sq_num = 0;
    for (int i = 0; i<n; i++){
        sq_sum += (x[i]-mean) * (x[i]-mean) }
    return sqrt(sq_sum /n-1); // returning Stdde
}
int main (){
    double x[] = { ----- elements ----- };
    int n = sizeof(x)/ sizeof(x[0]); // array size
    // Output and function call
    cout << "Standard Deviation: " << stdev (x,n) << endl;
    system ("pause"); // system process suspend
    return 0;
}
```

Output

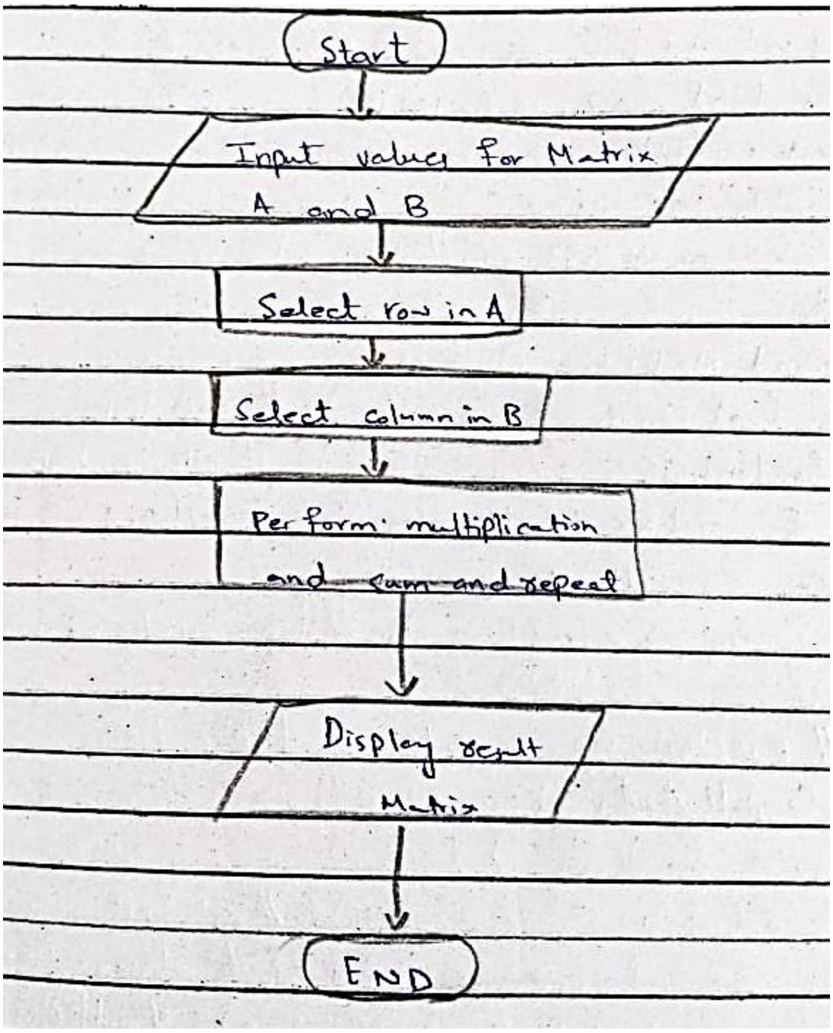
```
Standard Deviation: 2.69258
Press any key to continue . . .
```

Q3: write a program to multiply two 3x3 matrices .Make with function that can pass array to each function.

Block Diagram



Flowchart



Code

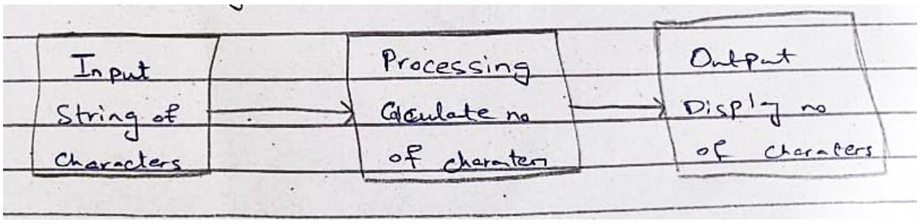
```
# include <iostream>
# include <stdlib.h>
using name space std;
// Function for input matrices
void GetValue ( int (&arrA)[3][3], int (&arrB)[3][3]){
    // First Matrix
    for (int i=0; i<3; i++){ // row loop
      for (int j=0; j<3; j++){ //column loop
      cout << "Enter value for first arr [" << i <<
        "][" << j << "]";
      cin >> arrA[i][j]; // user enter value
    }}
   // second Matrix.
    For (int i=0 ; i<3; i++){ // row loop
      for (int j=0 ; j<3 ; j++){ // column loop
        { cout << "Enter value for second arr["
      << i << "][" << j << "]:" ;
          cin >> arrB[i][j]; // User Input }
    }}
    // Function to compute Multiplication
 void Compute AxA (int arrA[][3], int arrB[][3]
      , int (&res)[3][3]) {
      for (int i=0; i<3; i++){ // row loop
         for (int j=0; j<3; j++){ // column loop
           for (int k=0; k<3; k++){ // multiplication loop
              res[i][j] += arrA[i][k] * arrB[k][j]; }
        }}}
//Function to print the array
void PrintArray (int res[][3]) {
    For (int i=0; i<3; i++) { //row loop
      For (int j=0; j<3; j++){ // column loop
            cout << res[i][j] << "\t";
      }
         cout << endl; //next row
      }}
int main (){
    int arr[3][3], arrB[3][3], res[3][3]={0} // arrays
    GetValue (arrA, arrB);  // Getting input
    Compute AxA (arrA, arrB, res); // Multiplying A and B
     Print Array (res); // results
    system ("pause");
    return 0;
  }
```

Output

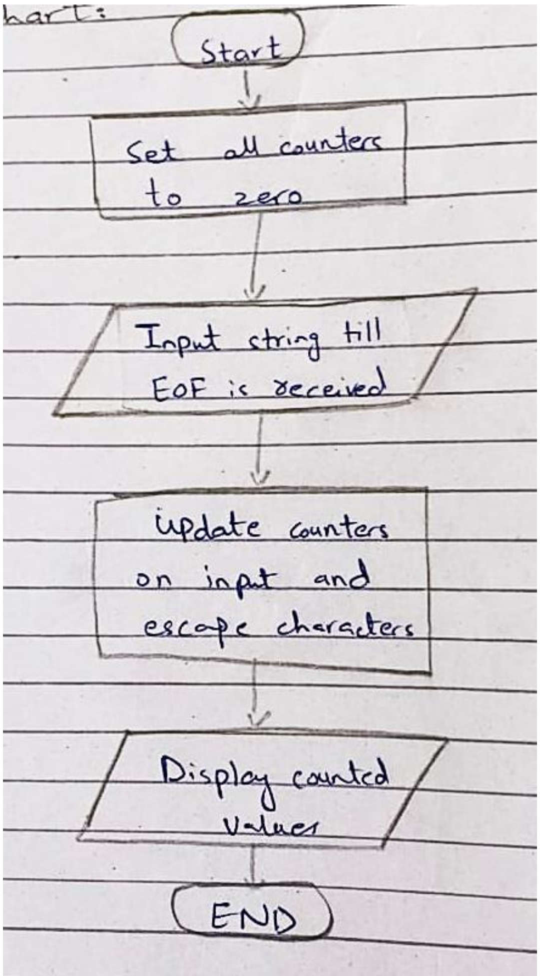
```
Enter value for first arr[0][0]: 1
Enter value for first arr[0][1]: 2
Enter value for first arr[0][2]: 3
Enter value for first arr[1][0]: 4
Enter value for first arr[1][1]: 5
Enter value for first arr[1][2]: 6
Enter value for first arr[2][0]: 7
Enter value for first arr[2][1]: 8
Enter value for first arr[2][2]: 9
Enter value for second arr[0][0]: 11
Enter value for second arr[0][1]: 12
Enter value for second arr[0][2]: 13
Enter value for second arr[1][0]: 14
Enter value for second arr[1][1]: 15
Enter value for second arr[1][2]: 16
Enter value for second arr[2][0]: 17
Enter value for second arr[2][1]: 18
Enter value for second arr[2][2]: 19
90      96      102
216     231     246
342     366     390
Press any key to continue . . .
```

Q4: Write a program that takes a string from the user. The program then calculates the Total number of Characters, Total number of Spaces, Total number of Tabs, Total number of Lines?

Block Diagram



Flowchart



Code

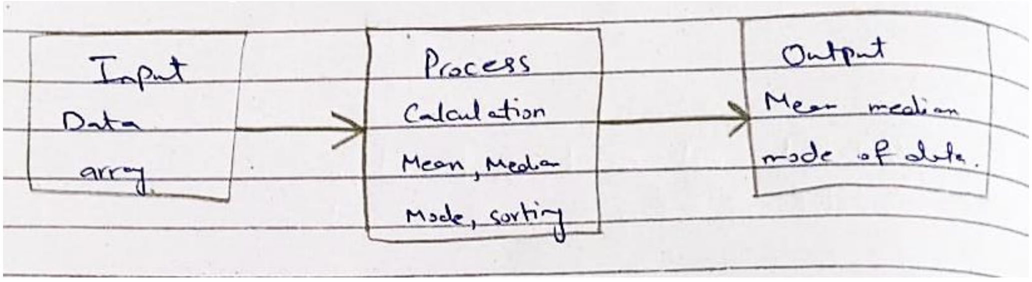
```
# include <iostream>
# include <stdlib.h>
using namespace std;
    // Input and counting Function
void input ( int &spc, int &tab, int &charc, &int lns)
  {
    string line = " "; // Input variable.

    Cout << " Enter text: ";
    while ( getline (cin, line)) { // TRUE until EOF
        For (char c : line) // For c in line
        {
          if ( c == ' ')
              spc ++
          else if ( c == '\t')
              tab ++;
            charc ++;     // character increment
        }
        lns++ ;    // line increament
      }
  }
int main () {
    Int spc = 0, tab =0 , charc = 0, lns = 0; // variables
    input (spc, tab, charc, lns); // call
   cout << "Spaces: " << spc << endl << "Tabs: " << endl
    << "Characters: " << charc << "Lines: " << lns; // output
    system ("pause");
    return 0;
 }
```

Output

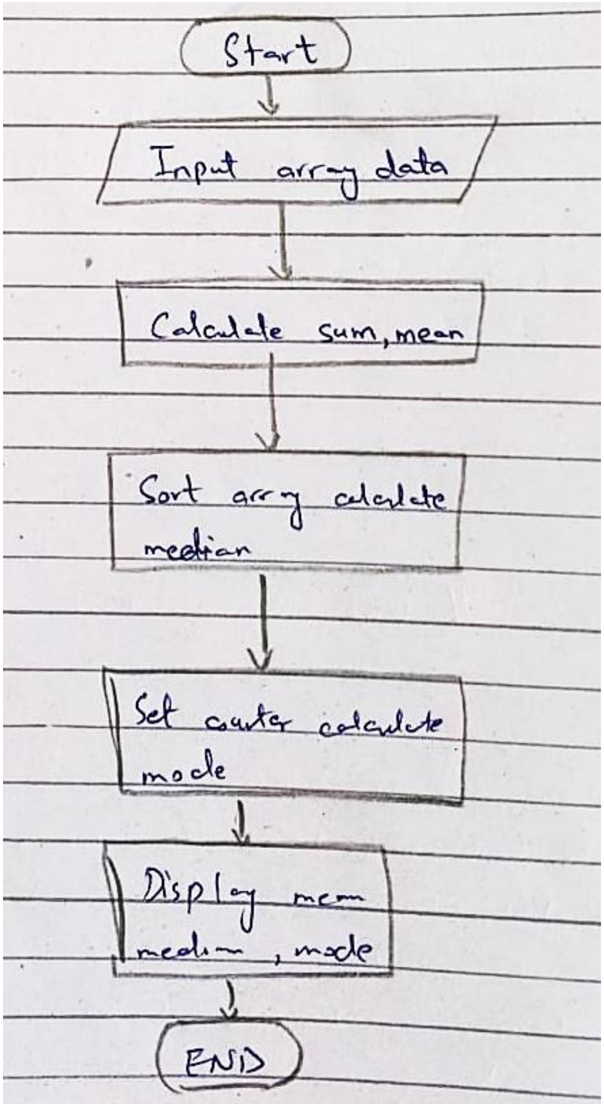
```
To Input Text press Ctrl + Z to generate EOF character in next line and hit return.
Enter text: Hello my name is Ashhad              Good morning
List
        1. Apples
        2. Cows
        3. Trucks
^Z
Spaces: 8
Tabs: 5
Characters: 69
Lines: 5
Press any key to continue . . .
```

Q5: Write a program to calculate mean, median and mode of an array using function

Block Diagram



Flowchart



Code

```
# include <iostream>
# include <stdlib.h>
# include <algorithm>   // for sort() func
using namespace std;

double mean (double arr [], int n){
    double sum = 0;      // Initiallize Sum
    for (int i = 0; i<n; i++){
        sum += arr[i]; }   // Summation
    return sum/n;          // Results returned
}

double median(double arr[], int n) {
    sort( arr, arr+n);          // Sorting array
    double median = 0;          // Initialize median
    if (n%2 == 0)               // If n is even
      median = (arr[n/2] + arr[n/2 -1]) /2
    else                        // If n is odd
      median = arr [n/2];
    return median;              // Results returned
}

double mode (double arr[], int n) {
    double mode = 0;            // Initialized variables
    int freq = 0;
    for (int i=0; i<n; i++) {   // Element loop
      int count =0;
      for (int j=0; j<n; j++) { // Comparison loop
          if (arr[i] == arr [j])
            count ++;
          if ( count > freq) { // If count is greater
              freq = count;    // update frequency
              mode = arr [i];  // and mode
          }
      }
      return mode;             // Results returned
    }

int main () {
    double arr [] = { --- elements --- };
    int n = size of (arr) / size of(arr[0]);
    // Function call and output
    cout << "Mean: " << mean (arr,n) <<endl;
    cout << "Median: " << median (arr,n) <<endl;
    cout << "Mode: " << mode (arr,n) << endl;

    system ("pause");
    return 0;
}
```

Output

```
Mean: 13.024
Median: 12
Mode: 15.5
Press any key to continue . . .
```